Notes: Plantin and Tirole (AER, 2018)

Marking to Market versus Taking to Market

Contents

1 Big picture

- **Question.** How should firms measure asset values for governance purposes when they can rely either on **market signals** from comparable assets (*marking to market*) or on the **realized price** obtained by actually selling the asset (*taking to market*)?
- **Setting.** The paper builds a dynamic agency model of corporate governance with three dates, insiders and outsiders, noisy public market signals, and a decentralized resale market with informed buyers. It first solves for the privately optimal contract, then endogenizes resale prices, liquidity, and the quality of market data.
- **Core challenge.** Fair-value accounting can use market data cheaply, but those data are noisy and may reward insiders for luck. Historical-cost-style accounting avoids that noise, but it may force firms to realize gains inefficiently, that is, to sell assets just to validate insiders' claims. The paper studies the optimal trade-off between these two distortions.
- **Main theoretical contribution.** The paper shows that the optimal contract can be implemented by an accounting rule with an endogenous **degree of conservatism**. This accounting rule determines how aggressively latent capital gains are recognized. The paper then studies how privately chosen conservatism interacts with the liquidity of secondary markets.
- **Main results.**
  - The privately optimal contract rewards insiders either when the public signal is sufficiently good or when the asset is actually resold above a reserve price.
  - This contract admits a natural accounting implementation: the more conservative the accounting rule, the less the firm recognizes unrealized gains and the more insiders are induced to **take the asset to market**.
  - With endogenous bids and **secret contracts**, there may be two equilibria: a **liquid, constrained-efficient equilibrium** and an **illiquid, unstable equilibrium**. In the illiquid equilibrium, firms rely too much on mediocre market data and sell too rarely at distressed prices.
  - A modest regulation that forces firms to adopt a **more conservative** accounting regime can eliminate the illiquid equilibrium.
  - With **posted contracts**, equilibrium becomes unique, but firms compete for liquidity by lowering reserve prices, which raises agency costs when liquidity is exogenous.
  - With **endogenous liquidity**, laissez-faire is generically inefficient because firms rely excessively on market-value information generated by *other firms'* sales and too little on the realization of *their own* capital gains.
  - Both liquidity support and tighter accounting conservatism can improve welfare because they thicken the resale market and improve the informativeness of prices.
- **Why it matters.** Accounting is not just a passive reporting technology. It shapes managerial incentives, asset-market liquidity, and the quality of information that firms use to govern themselves.
- **Policy implication.** The paper provides an efficiency rationale for **more conservative accounting** than firms privately choose under laissez-faire, especially when market signals themselves depend on the volume and quality of other firms' trades.

- **Economic takeaway.** A firm's accounting rule affects not only its own contract but also the liquidity and informational content of the market for similar assets. Because firms do not internalize these externalities, laissez-faire leads to **too much fair value** and **too little taking to market**.

2 Data and measurement (key objects)

2.1 This is a theory paper: the key objects are model primitives

- There is **no empirical dataset**. The paper is entirely theoretical.
- The important objects are:
  - the insider–outsider agency problem,
  - the public signal about asset value,
  - the distribution of resale bids,
  - the accounting rule mapping signals into book values,
  - market liquidity, measured by the mass of informed buyers and the ease of resale.

2.2 Dates, parties, and interpretation

- **Dates.** There are three dates, 0, 1, and 2.
- **Principal.** The principal represents outsiders: diffuse shareholders, arm's-length creditors, or a prudential supervisor.
- **Agent.** The agent represents insiders: managers, controlling shareholders, or directors.
- **Key interpretation of the date-1 transfer.** The utility transfer to the agent at date 1 can be interpreted as:
  - continuation or expansion versus liquidation or downsizing,
  - transfer of corporate control,
  - managerial compensation.

2.3 Project structure and agency problem

- The agent chooses between two project types at date 0.
- At date 2, one project type pays h and the other pays l, with $h > l$.
- The agent receives a private signal about which project is better.
- If the agent **behaves**, the signal is correct with probability p.
- If the agent **shirks**, the signal is correct with probability $p - \Delta p$ and the agent receives a private benefit B.

- Define

$$\beta \equiv \frac{B}{\Delta p} \leq 1.$$

This is the utility rent needed to induce effort per unit increase in signal precision.

2.4 Measurement technology 1: public signal

- At date 1, outsiders observe a public signal $s \in \mathbb{R}$.
- Conditional on final payoff $y \in \{h, l\}$, the signal has density $f_y(s)$ and distribution $F_y(s)$.
- The model assumes the likelihood ratio f_h/f_l is strictly increasing, so higher signals are more indicative of high value.
- Interpretation: the signal is market data from comparable assets, public quotes, or transaction data that reveal something about what the firm's asset would fetch.

2.5 Measurement technology 2: resale opportunities

- The firm can also **sell the asset** at date 1.
- There are uninformed buyers who submit bids up to l, and informed buyers who may bid above l when they know the asset is good.
- Let H be the distribution of the highest informed bid for a good asset.
- With probability q_0, no informed buyer is matched to the firm, so there is no informative high bid.
- The support of informed bids for a good asset is $[r^-, r^+]$, with $r^- > l$ and $r^+ \leq h$.

2.6 Accounting measure and conservatism

- The paper defines an accounting measure as:
  - the realized resale proceeds if the asset is sold,
  - otherwise a signal-based book value $m(s)$.
- The key parameter is the **degree of conservatism** α:
  - low α means unrealized gains are recognized aggressively,
  - high α means book value responds only gradually to favorable market news.
- In the limit $\alpha \geq 1 - q_0$, the book value is always l absent a sale. This is the historical-cost limit: only realized gains matter.

2.7 Short-term equilibrium objects

- There is a continuum of firms and a mass λ of informed buyers.
- Buyers are randomly matched to firms.

- If matching is urn-ball, the probability that a firm matches with k informed buyers is

$$q_k(\lambda) = \frac{\lambda^k e^{-\lambda}}{k!}.$$

- The key objects are:
  - $q_0(\lambda)$: probability no informed buyer arrives,
  - $q_1(\lambda)$: probability exactly one informed buyer arrives,
  - λ: market liquidity,
  - H: highest-bid distribution.

2.8 Long-term equilibrium objects

- In the long-term equilibrium, the mass λ of informed buyers is endogenous.
- Buyers pay an information-acquisition cost κ to become informed.
- The quality of public signals also depends on market liquidity, so the public signal distribution becomes $F_y(s, \lambda)$.
- This creates two externalities:
  - more informed buyers bid more aggressively and reduce resale discounts,
  - more informed buyers generate more informative market data for all firms.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline agency model and second-best benchmark

The agent must be given incentives to choose the better project. If the principal could directly observe the date-2 payoff at date 1, the agent could be induced to behave by paying utility only when the payoff is high. The associated second-best cost is

$$p\beta.$$

The model's central question is how much extra agency cost is created by the **measurement friction**: at date 1 the principal does not observe the final payoff directly and must instead rely on signals and possible resale prices.

Interpretation.

- Without a measurement problem, incentives are straightforward.
- With a measurement problem, the principal must choose how much to rely on noisy public signals versus costly realized sale prices.

3.2 Optimal contract with exogenous bids

Let

$$\bar{\beta} \equiv \sup_{s} \{F_l(s) - q_0 F_h(s)\}. \tag{1}$$

The optimal contract is characterized in Proposition 1. If $\beta > \bar{\beta}$, high effort cannot be elicited. If $\beta \le \bar{\beta}$, the optimal contract is summarized by a signal threshold σ and a reserve price r.

The agent is rewarded if:

- the public signal is above σ, or
- the public signal is below σ but the asset is sold above the reserve price r.

The agency cost of the optimal contract is

$$p\beta + 1 - F_l(\sigma) + pF_h(\sigma) \int_r^h (h-t)\, dH(t). \tag{2}$$

The principal chooses (σ, r) to solve

$$\min_{\{\sigma, r\}} \left\{ (1-p)(1-F_l(\sigma)) + p(1-F_h(\sigma)) + pF_h(\sigma) \left[1 - H(r) + \int_r^h (h-t)\, dH(t) \right] \right\} \tag{3}$$

subject to

$$F_l(\sigma) - H(r)F_h(\sigma) = \beta, \tag{4}$$

and

$$r^- \le r \le r^+. \tag{5}$$

Ignoring the bounds on r, the first-order condition is

$$\frac{f_h(\sigma)}{f_l(\sigma)} = \frac{p(h-r)+1}{p\int_r^h H(t)\, dt}. \tag{6}$$

Interpretation.

- Equation (4) is the incentive-compatibility constraint: the expected gain to the agent from exerting effort must cover the private benefit of shirking.
- Equation (2) decomposes agency costs into three pieces:
  - $p\beta$: the second-best incentive cost,
  - $1 - F_l(\sigma)$: the cost of **rewarding luck** when the public signal is favorable even though the asset is bad,
  - $pF_h(\sigma) \int_r^h (h-t)\, dH(t)$: the expected cost of **taking to market** at a discount.
- Equation (6) says the optimal contract equalizes the marginal cost of relying on the signal (*marking to market*) and the marginal cost of relying on resale (*taking to market*).

3.3 Implementation with an accounting measure

After observing signal s, define the probability that the resale price would exceed $x > l$ by

$$D_s(x) = \frac{p\, f_h(s)\big(1 - H(x)\big)}{p\, f_h(s) + (1-p)\, f_l(s)}. \tag{7}$$

For each degree of conservatism $\alpha \in (0, 1)$, define the accounting measure

$$m_\alpha(s) = \inf\{x > l \mid D_s(x) < \alpha\}. \tag{8}$$

Proposition 2 shows that the optimal contract (σ, r) can be implemented with an accounting rule whose degree of conservatism is

$$\alpha = \frac{p\, f_h(\sigma)\big[1 - H(r)\big]}{p\, f_h(\sigma) + (1-p)\, f_l(\sigma)}. \tag{9}$$

Interpretation.

- $m_\alpha(s)$ is a signal-based book value consistent with market resale values.
- Lower α means the book value quickly reflects possible latent gains.
- Higher α means recognition of latent gains is more conservative.
- Historical-cost accounting appears as the limiting case where unrealized gains are essentially ignored unless the asset is actually sold.

3.4 Short-term equilibrium with endogenous bids

With a mass λ of informed buyers, urn-ball matching implies

$$q_k(\lambda) = \frac{\lambda^k e^{-\lambda}}{k!}. \tag{10}$$

In a **high-winner-curse** equilibrium, uninformed buyers optimally never bid above l, and informed buyers with good news bid in a way that does not depend on the public signal.

A key implication is that the probability a good asset fails to sell is exactly the probability that no informed buyer is matched, so

$$H(r) = q_0. \tag{11}$$

Then the incentive constraint becomes

$$F_l(\sigma) - q_0 F_h(\sigma) = \beta, \tag{12}$$

and the optimality condition becomes

$$\frac{f_h(\sigma) q_0}{f_l(\sigma)} = \frac{1 + \frac{1}{p(h-r)}}{1 + \frac{\lambda q_1}{q_0(1-q_0)}}. \tag{13}$$

These are the two equations that pin down the equilibrium pair (σ, r).

Interpretation.

- The left-hand side captures how informative the public signal is near the cutoff.
- The right-hand side captures how costly it is to rely on resale rather than on signals once informed buyers endogenously bid.
- The more liquid the market is, the less painful it is to rely on realized sales.

3.5 Multiplicity of equilibria under secret contracts

Proposition 3 shows that there is always a **liquid** equilibrium $(\bar{\sigma}, \bar{r})$ with

$$f_h(\bar{\sigma})q_0 \geq f_l(\bar{\sigma}),$$

and there may also exist an **illiquid** equilibrium $(\underline{\sigma}, \underline{r})$ with

$$f_h(\underline{\sigma})q_0 \leq f_l(\underline{\sigma}).$$

If $\beta \leq 1 - q_0$, the liquid equilibrium coincides with the second-best corner case in which the good asset is sold at price h with probability $\beta/(1 - q_0)$.

Interpretation.

- In the **liquid equilibrium**, buyers expect aggressive reserve prices, bid more aggressively, and firms optimally rely on higher reserve prices and more conservative accounting.
- In the **illiquid equilibrium**, buyers expect low reserve prices, bid less aggressively, and firms then rationally rely more on noisy market data and accept lower sale prices.
- The multiplicity comes from the feedback between **beliefs about reserve prices** and **optimal bidding**.

3.6 Regulating marking to market

Proposition 4 studies a regulation that forbids firms from rewarding agents solely on the basis of low-quality market signals below some threshold σ'.

Result.

- The illiquid equilibrium is **unstable**: a small increase in the cutoff pushes the economy toward the liquid equilibrium.
- The liquid equilibrium is **constrained efficient**: making the cutoff even higher destroys incentive compatibility.

Under the accounting implementation, this is equivalent to requiring a **higher degree of conservatism**:

$$\alpha(\sigma) = \frac{p\, f_h(\sigma)(1 - q_0)}{p\, f_h(\sigma) + (1 - p)\, f_l(\sigma)}. \tag{14}$$

Interpretation.

- A small move toward historical-cost-style measurement can coordinate firms on the efficient equilibrium.

- The paper's policy message in this section is not "ban fair value," but rather "prevent excessively aggressive recognition of latent gains."

3.7 Posted contracts and competition for liquidity

The paper next allows firms to post contracts publicly. Buyers can then direct their matching toward firms.

Proposition 5:

- equilibrium becomes **unique**,
- all firms post the same contract,
- buyers match uniformly across firms,
- if $\beta > 1 - q_0$, firms post $(\bar{\sigma}, r^P)$ with $r^P < \bar{r}$.

Interpretation.

- Public commitment eliminates the bootstrap-style multiplicity created by secret contracts.
- But firms now compete for liquidity, which induces them to post lower reserve prices than is efficient when liquidity is fixed.
- So uniqueness does *not* automatically imply efficiency.

3.8 Uncertain liquidity and crisis episodes

The paper also studies the case in which firms must write contracts before observing the market liquidity state λ.

Proposition 6 shows:

- the reserve price r is **constant** across liquidity states,
- the signal threshold $\sigma(\lambda)$ is **increasing in liquidity**.

Interpretation.

- When markets are liquid, resale is relatively cheap, so the firm can afford to rely less on the public signal and ask insiders to realize gains more often.
- When markets are illiquid, it is efficient to rely less on resale and more on accounting measures that do not force fire sales.
- The authors connect this logic to the temporary suspension of marking-to-market rules during the 2008 financial crisis.

3.9 Long-term equilibrium with endogenous liquidity

In the full equilibrium, buyers decide whether to become informed by paying a cost κ. The public signal quality also depends on the amount of informed trading.

The paper assumes

$$\frac{\partial F_h(s,\lambda)}{\partial \lambda} \le 0, \tag{15}$$

and

$$f_l(s,\lambda)\frac{\partial F_h(s,\lambda)}{\partial \lambda} \le f_h(s,\lambda)\frac{\partial F_l(s,\lambda)}{\partial \lambda}. \tag{16}$$

These conditions mean that more informed trading improves the signal, especially for high-value assets.

A stable high-winner-curse equilibrium with endogenous liquidity solves

$$\frac{f_h(\sigma,\lambda)q_0(\lambda)}{f_l(\sigma,\lambda)} = \frac{1+\frac{1}{p(h-r)}}{1+\frac{\lambda q_1(\lambda)}{q_0(\lambda)\big(1-q_0(\lambda)\big)}} > 1, \tag{17}$$

$$F_l(\sigma,\lambda) - F_h(\sigma,\lambda)q_0(\lambda) = \beta, \tag{18}$$

$$pF_h(\sigma,\lambda)\frac{q_1(\lambda)(h-r)}{1-q_0(\lambda)} = \kappa. \tag{19}$$

Proposition 7 shows such an equilibrium exists for sufficiently low κ, and the equilibrium agency costs are

$$p\beta + 1 - F_l(\bar{\sigma}(\lambda),\lambda) + \lambda\kappa. \tag{20}$$

Interpretation.

- Equation (19) is the free-entry condition: informed buyers enter until expected informational rents equal information-acquisition costs.
- More informative markets make resale easier and improve the signal. This is the key feedback absent from the partial-equilibrium analysis.

3.10 Welfare and policy in the long-term equilibrium

To the equilibrium (σ, r, λ) corresponds a degree of conservatism

$$\alpha = \frac{p\, f_h(\sigma)\big[1-q_0(\lambda)\big]}{p\, f_h(\sigma) + (1-p)\, f_l(\sigma)}. \tag{21}$$

Proposition 8 shows that two interventions locally reduce agency costs:

- **liquidity support**, such as subsidizing information acquisition,
- **higher accounting conservatism**.

Interpretation.

- Laissez-faire produces **too little liquidity**.
- Firms do not internalize two positive externalities they create for other firms:
  1. more informed buyers bid more aggressively, lowering resale costs,

2. more informed buyers generate better market data.

- Therefore firms rely too much on transactions by *other* firms rather than generating information through their *own* trades.

Proposition 9 then studies the case of **posted contracts with endogenous liquidity**. Posted contracts internalize the first externality (competition for resale liquidity), but not the second (better market data). As a result, equilibrium remains locally inefficient whenever the signal-quality externality is strict.

4 Identification and instruments (what makes the estimates credible)

4.1 What the standard accounting debate misses

- The usual debate compares **fair value** and **historical cost** as if they were static reporting rules.
- This paper shows that the right comparison is **equilibrium**: accounting rules change the liquidity of asset markets, and that liquidity changes the informativeness of market prices.
- Therefore one cannot judge the desirability of fair value while holding market liquidity fixed.

4.2 What pins down the privately optimal contract

- The optimal contract is pinned down by two measurable objects inside the model:
  - how informative the public signal is near the threshold, captured by f_h/f_l,
  - how costly resale is, captured by the reserve price and bid distribution H.
- The contract trades off two distortions:
  - **reward for luck** from relying on noisy market data,
  - **gains trading** from forcing resale to validate latent gains.

4.3 What generates equilibrium multiplicity

- Under secret contracts, buyers do not observe a firm's chosen reserve price.
- They bid based on beliefs about contracts.
- Firms, in turn, choose contracts based on beliefs about bids.
- This creates a self-fulfilling feedback:
  - pessimistic bidding beliefs justify low reserve prices and illiquid outcomes,
  - optimistic bidding beliefs justify high reserve prices and liquid outcomes.

4.4 What creates inefficiency in the long run

- The key inefficiency is an **informational externality**.
- A firm's willingness to sell affects:
    1. the rents and aggressiveness of informed buyers in the market for similar assets,
    2. the informational content of future public signals observed by other firms.
- Individual firms do not internalize these effects, so laissez-faire features **too little liquidity** and **too little conservatism**.

4.5 Why the theory is robust

- The paper shows the main mechanism survives a number of extensions:
    - posted contracts,
    - uncertain liquidity states,
    - agents who value date-2 consumption,
    - alternative welfare criteria,
    - false positive and false negative mistakes by informed buyers.
- These extensions change some comparative statics, but not the core message that accounting choices and liquidity are jointly determined.

4.6 Relation to the literature

- **Agency theory.** The paper is rooted in Holmström-style performance measurement, but adds an explicit trade-off between a free noisy signal and a costly precise signal obtained by resale.
- **Accounting theory.** It contributes to the literature on the real effects of accounting by endogenizing both governance contracts and market liquidity.
- **Market microstructure and thick-market externalities.** The model is related to competitive search and thick-market externalities, but here the externality runs through firms' accounting choices.
- **Fire sales and market monitoring.** The paper also connects to settings where resale prices matter for governance or regulatory constraints.

5 Tables and figures

5.1 Figure 1: Timeline

- Figure 1 lays out the timing:
    - at date 0 the principal and agent sign a contract and the agent selects a project,

- at date 1 the public signal is realized, buyers submit bids, fake bids may be added, and the asset may be sold,
- at date 2 the asset payoff is realized.

- The figure is useful because the entire paper is about a **date-1 measurement problem**: rewards must be based on information available before final cash flows are realized.

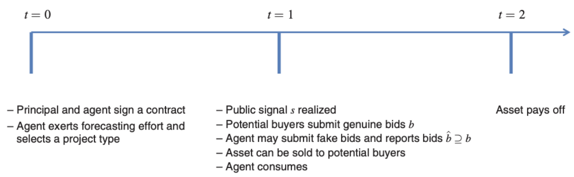

FIGURE 1. TIMELINE

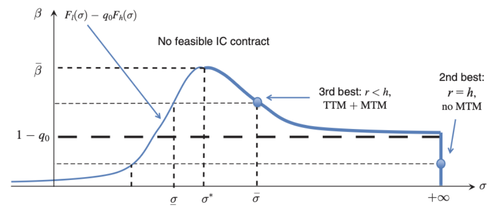

FIGURE 2. EQUILIBRIA FOR THREE VALUES OF β

5.2 Figure 2: Equilibria for three values of β

- Figure 2 is the key diagram of the paper.
- The horizontal axis is the signal threshold σ; the relevant curve is $F_l(\sigma) - q_0 F_h(\sigma)$.
- For low enough β, the second-best can be sustained without marking to market.
- For intermediate β, there can be **two equilibria**:
  - a low-σ illiquid equilibrium with low reserve prices and heavy reliance on market data,
  - a high-σ liquid equilibrium with high reserve prices and greater reliance on realized sales.
- For high β, no feasible incentive-compatible contract exists.
- Read the figure as showing how the severity of the incentive problem determines whether the economy ends up in a liquid or illiquid governance-liquidity configuration.

5.3 No empirical tables: the evidence is in the proposition sequence

- This paper has no empirical tables.
- The quantitative force of the argument comes from the sequence of propositions:
  - Proposition 1 and 2: privately optimal contract and accounting implementation,
  - Proposition 3 and 4: multiplicity and regulation under secret contracts,
  - Proposition 5 and 6: posted contracts and uncertain liquidity,
  - Proposition 7 to 9: endogenous liquidity, inefficiency, and policy.
- In other words, the paper's "evidence" is theoretical: it shows step by step how equilibrium feedback overturns the naive view that more fair value is always better.

6 Short summary

- **Method.** Embed accounting measurement into an agency model and then into a market equilibrium with informed trading and endogenous liquidity.
- **Core mechanism.** Conservative accounting induces firms to realize gains more often; those realizations create liquidity and improve price informativeness for everyone else.
- **Main equilibrium result.** Laissez-faire leads firms to rely excessively on other firms' transaction prices and insufficiently on their own realized sale prices.
- **Policy result.** More conservative accounting rules or liquidity support can improve welfare by thickening markets and improving market signals.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that the design of accounting rules cannot be separated from the equilibrium liquidity of asset markets.
- The privately optimal contract blends *marking to market* and *taking to market*, and this blend can be implemented by an accounting rule with an endogenous degree of conservatism.
- In equilibrium, firms choose accounting rules that are too aggressive from the social point of view.

7.2 Economic intuition

- Firms like fair-value-style measurement because it is cheap: it lets them reward insiders based on publicly available market data.
- But if every firm relies on other firms' prices and does not bring its own assets to market, then markets become thinner and prices become less informative.
- Historical-cost-style discipline has a hidden virtue: by forcing firms to realize gains, it generates the trading activity that creates useful price information.

7.3 Takeaway

This paper's central lesson is that there can be **too much fair value**. A firm privately prefers to contract on the information generated by other firms' transactions, but when all firms do this, the market for similar assets becomes less liquid and less informative. A socially better accounting regime is therefore **more conservative** than what laissez-faire delivers.